

Problem 1.

- a) See solution document.
- b) $a = \frac{F}{m} (\cos \theta + \mu_k \sin \theta) - \mu_k g$
- c) $F = \frac{\mu_k m g}{(\cos \theta + \mu_k \sin \theta)}$
- d) $\theta = \tan^{-1}(\mu_k)$
- e) Problem 2.
- f) C.

Problem 3.

B, C, E, F.

Problem 4.

D.

Problem 5.

- a) $a = g \frac{M \sin \theta - m}{m + M}$
- b) $a = g \frac{M(\sin \theta - \mu_k \cos \theta) - m}{m + M}$
- c)

Problem 6.

- a) C.

Problem 7.

- a) C.
- b) C.

Problem 8.

- a) A.